Telecom Churn Analysis

using SQL, Machine Learning & PowerBI


Juie Hedau

Introduction

Customer churn is a major challenge in the telecom industry, impacting revenue and growth.
This project analyzes customer data to identify churn patterns, visualize insights using Power BI, and predict churners using Python machine learning models, helping businesses improve customer retention.

Problem Statement

Telecom companies face high customer churn, leading to significant revenue loss.

There is a need for a data-driven approach to identify key churn factors and predict at-risk customers, enabling proactive retention strategies.

Objectives

- Analyze customer demographics, services, and account details to detect churn patterns.
- Build interactive dashboards in Power BI for churn monitoring.
- Develop a machine learning model to predict potential churners.
- Identify key drivers influencing churn (contracts, payments, services).
- Provide actionable recommendations to improve customer retention.

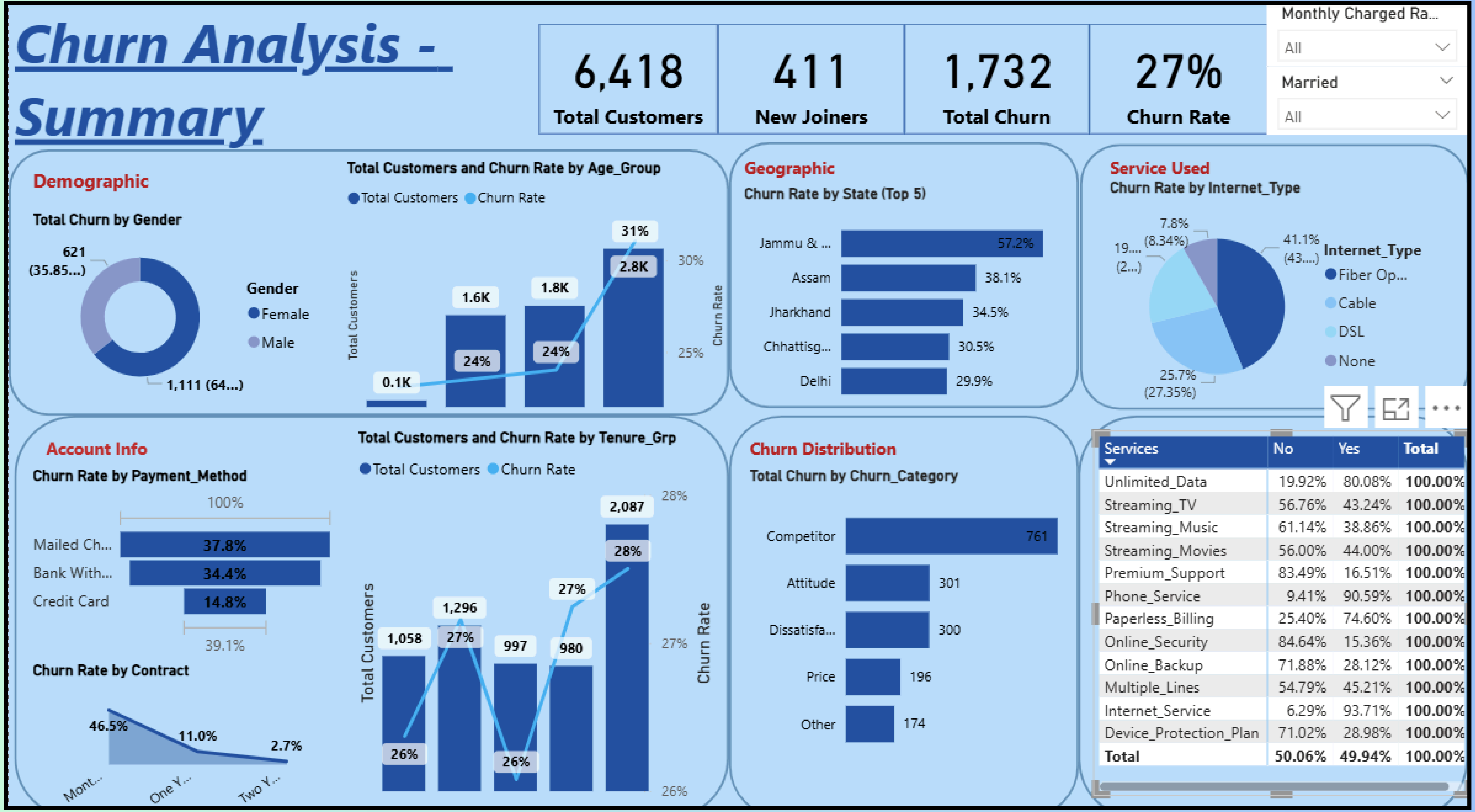
Dataset Overview

Source: Telecom Customer Dataset (public dataset /company-provided).

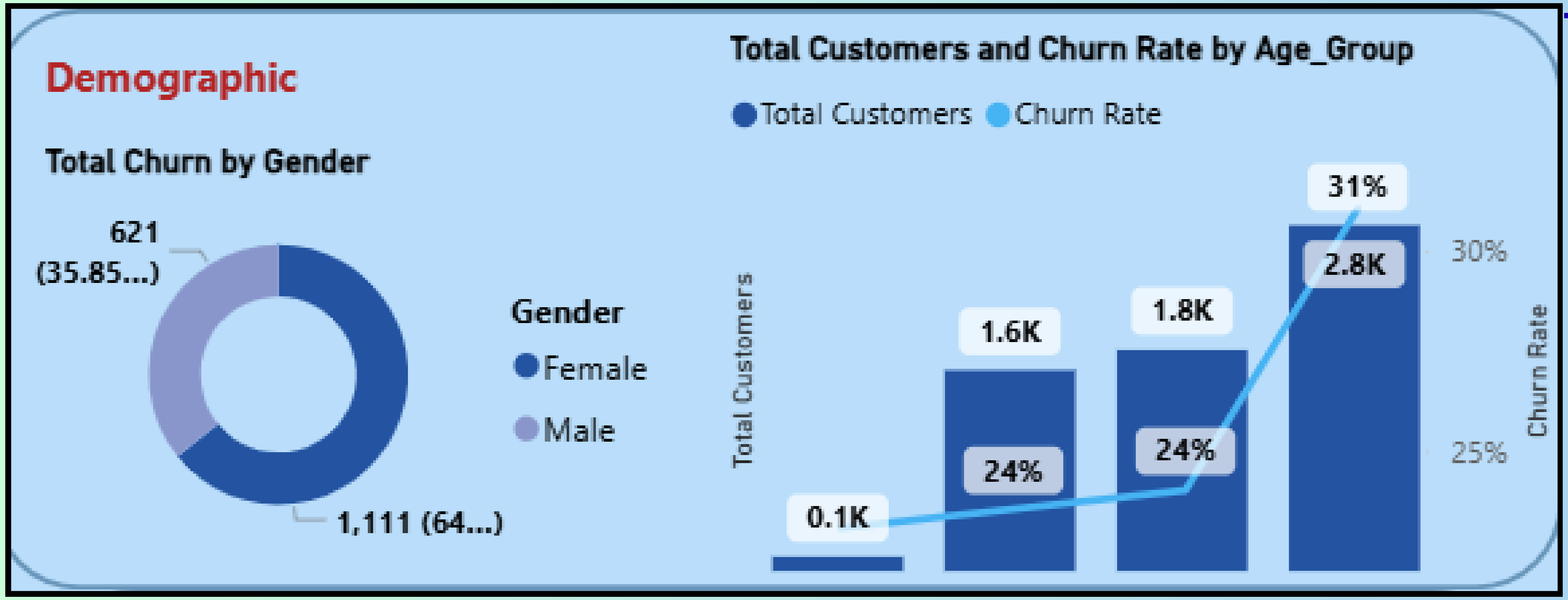
Size: ~7,000–8,000 customer records, 20+ features.

- **Key Features:**
- Customer Information: Customer ID, Gender, Age, Geography.
- Services Used: Internet Service, Streaming, Phone, Tech Support.
- Account Details: Contract Type, Tenure, Payment Method, Monthly Charges, Total Charges.
- Target Variable: Churn (Yes/No) – whether the customer left the company.

Churn Analysis Summary



Churned Rate - Demographic



Total Customers & Churn Rate by Age Group

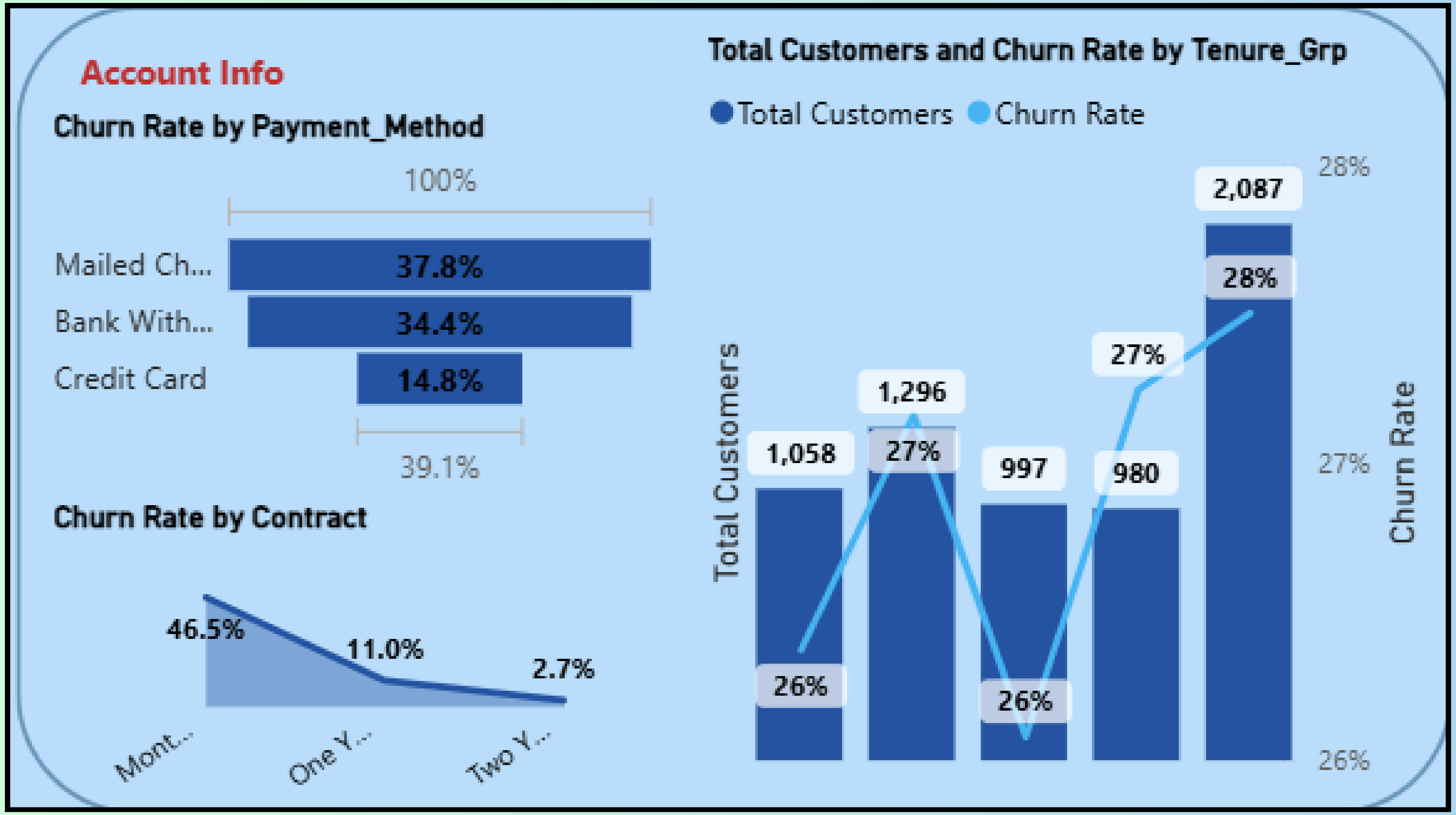
- <20 years: Very few customers (~100), churn rate ~24%.
- 20–35 years: 1.6K customers, churn rate ~24%.
- 35–50 years: 1.8K customers, churn rate ~24%.
- >50 years: Largest group (2.8K customers), churn rate 31%.

Total Churn by Gender

Out of total churned customers:

- Male customers contribute 64% (1,111 customers).
- Female customers contribute 36% (621 customers).

Churned Rate - Account Info



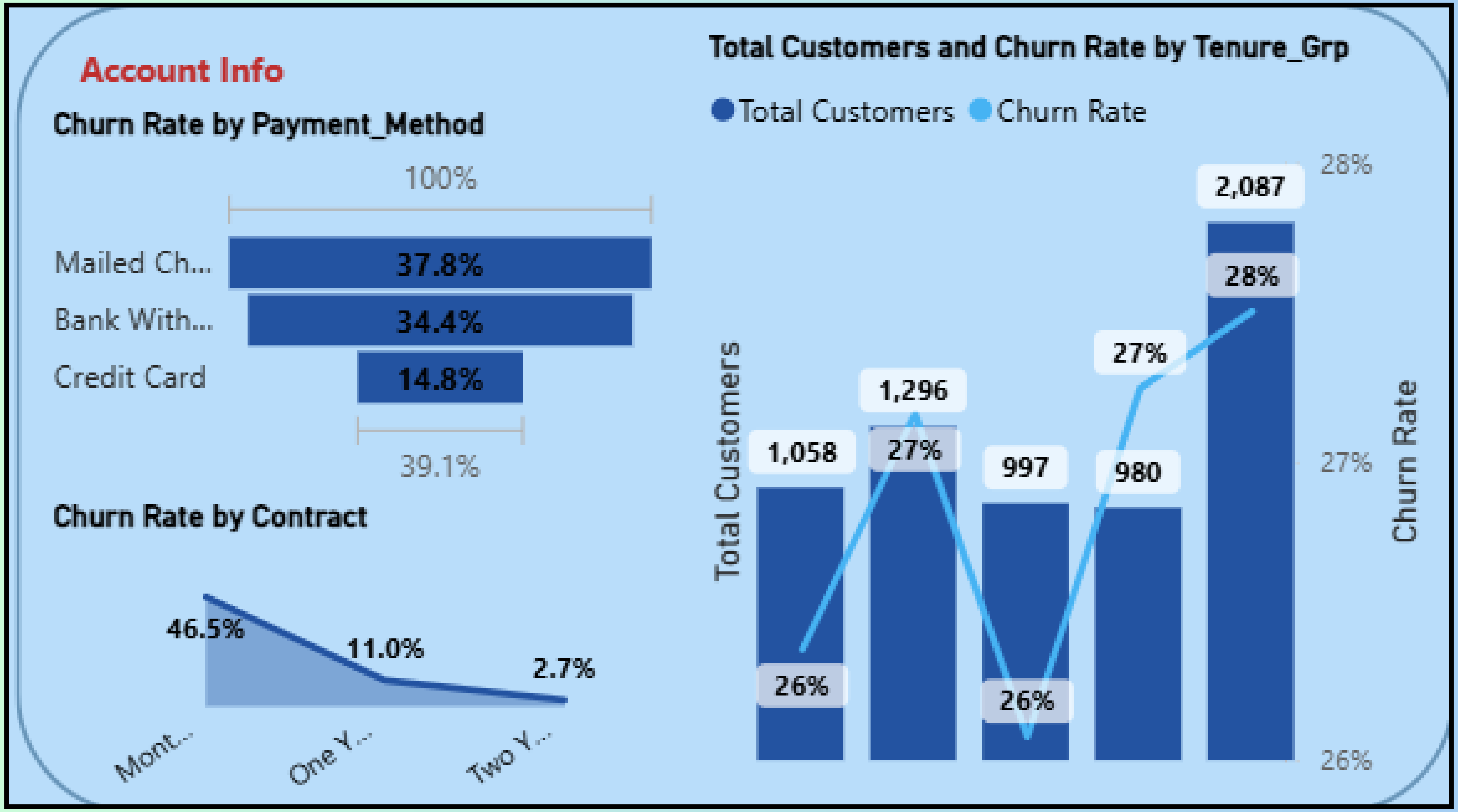
Total Customers & Churn by Tenure Group (Right)

- Customers with shorter tenure (<12 months) have churn rates around 26–27%.
- Customers with >24 months tenure have the highest churn rate (28%) despite being loyal for longer.
- **Insight**: Both new customers and some long-tenure customers (>2 years) are at risk of churn.

Churn Rate by Payment Method (Top-Left)

- Mailed Check customers have the highest churn rate (37.8%).
- Bank Withdrawal method also shows high churn (34.4%).
- Credit Card customers are the most loyal with a low churn rate (14.8%).

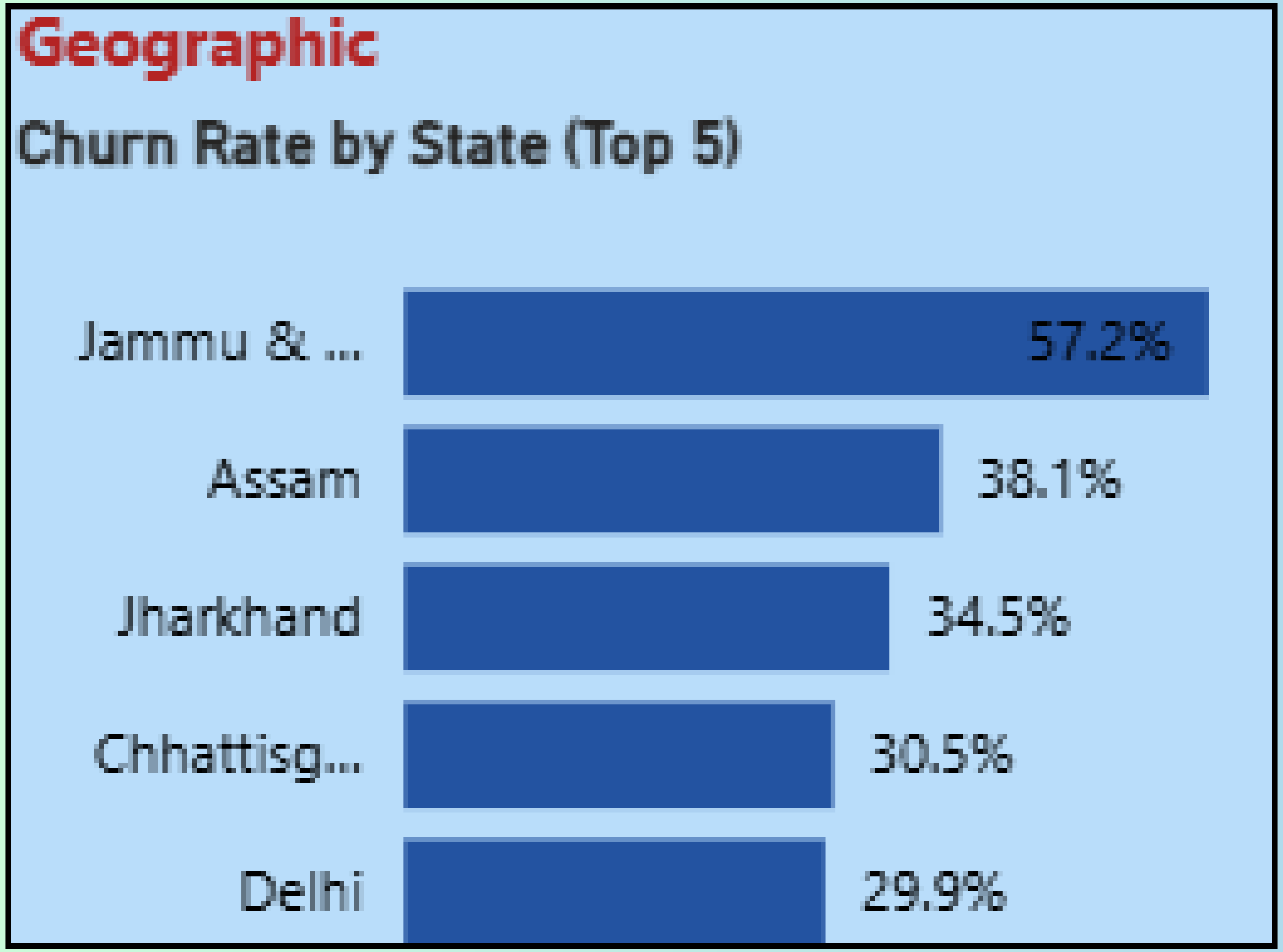
Churned Rate - Account Info



Churn Rate by Contract (Bottom-Left)

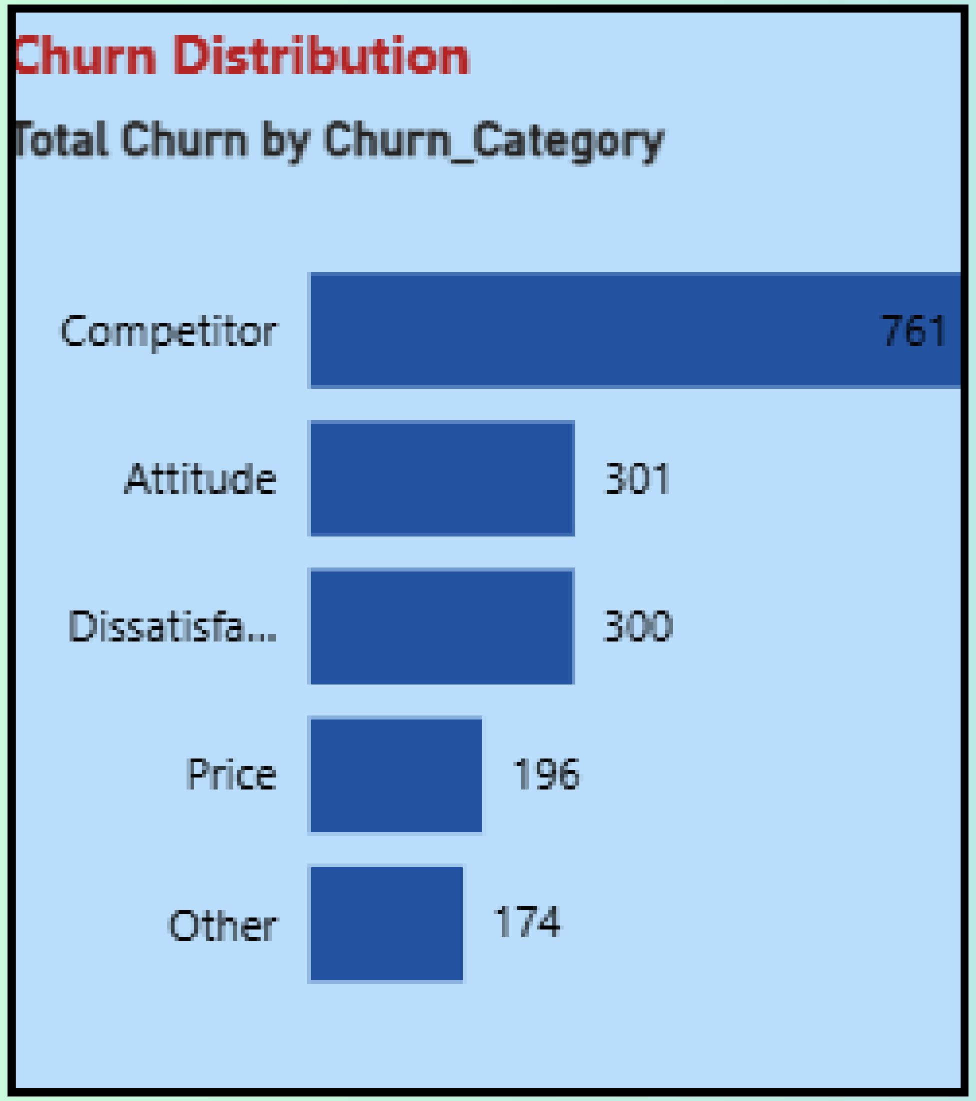
- Month-to-Month contracts: Very high churn (46.5%).
- One-Year contracts: Moderate churn (11%).
- Two-Year contracts: Very low churn (2.7%).

Churned Rate - Geographic

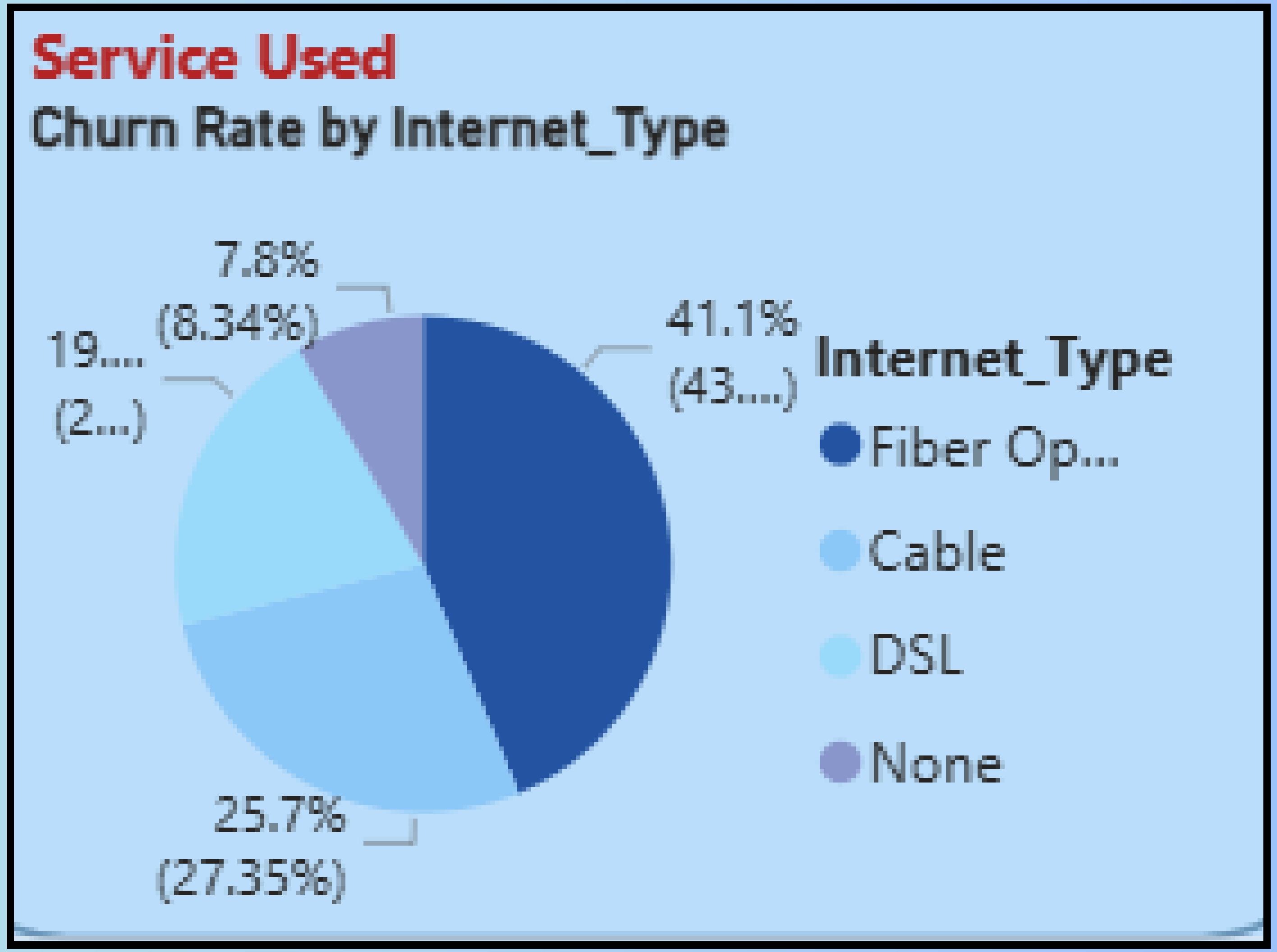


- Jammu & Kashmir has the highest churn rate (57.2%), showing a serious retention issue.
- Assam follows with a churn rate of 38.1%.
- Jharkhand (34.5%) and Chhattisgarh (30.5%) also show relatively high churn.
- Delhi has the lowest among the top 5 but still significant at 29.9%.

Based on-



- The customer churned due to-
- Highest no.of competitor (761)
- Attitude Behaviour (301)
- Dissatisfaction rate
- High Price
- other

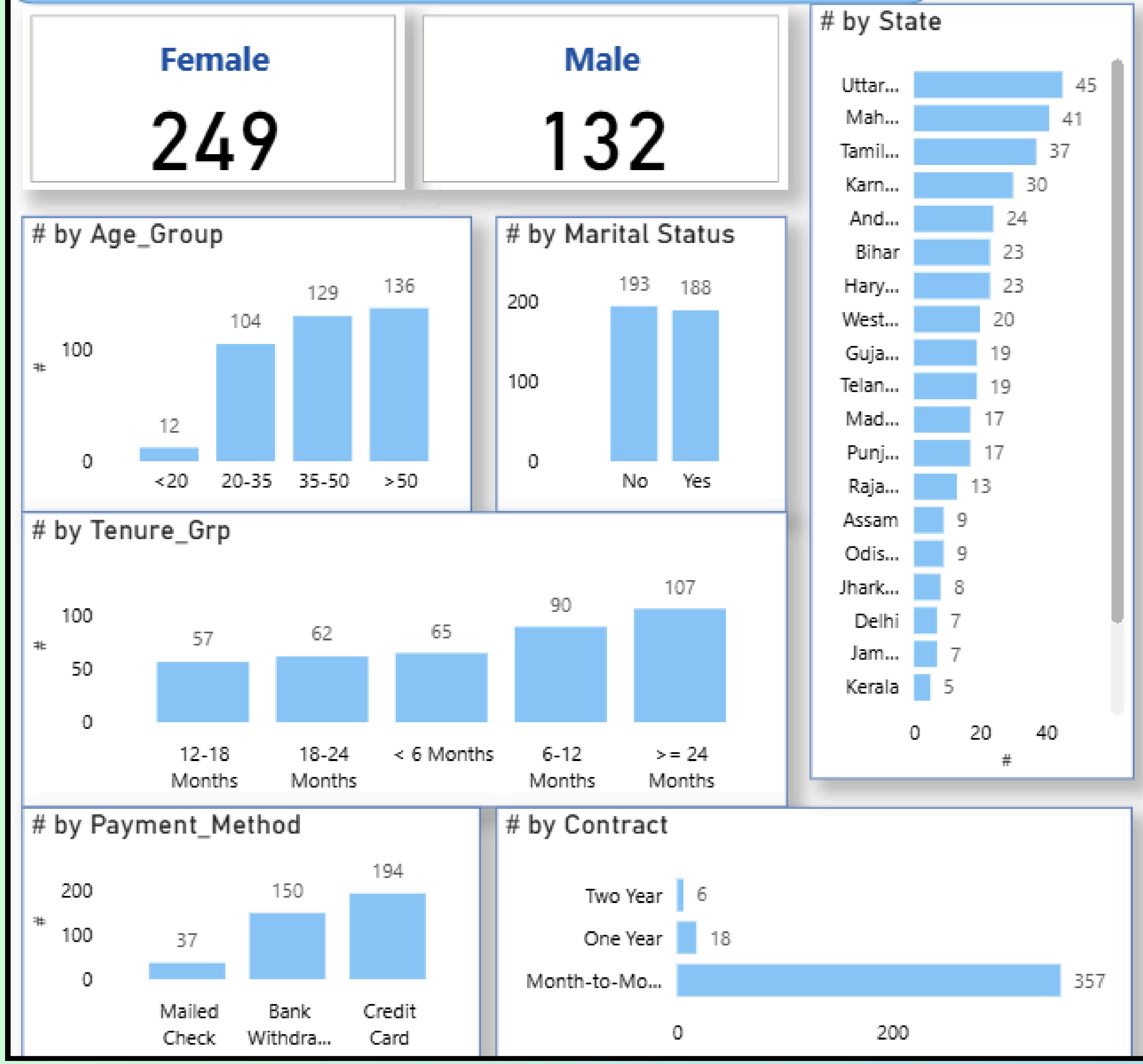


- Churn is highest among Fiber Optic users at about 41% of churners, followed by Cable at roughly 26%, DSL at about 25%, and None at about 8%.

Prediction

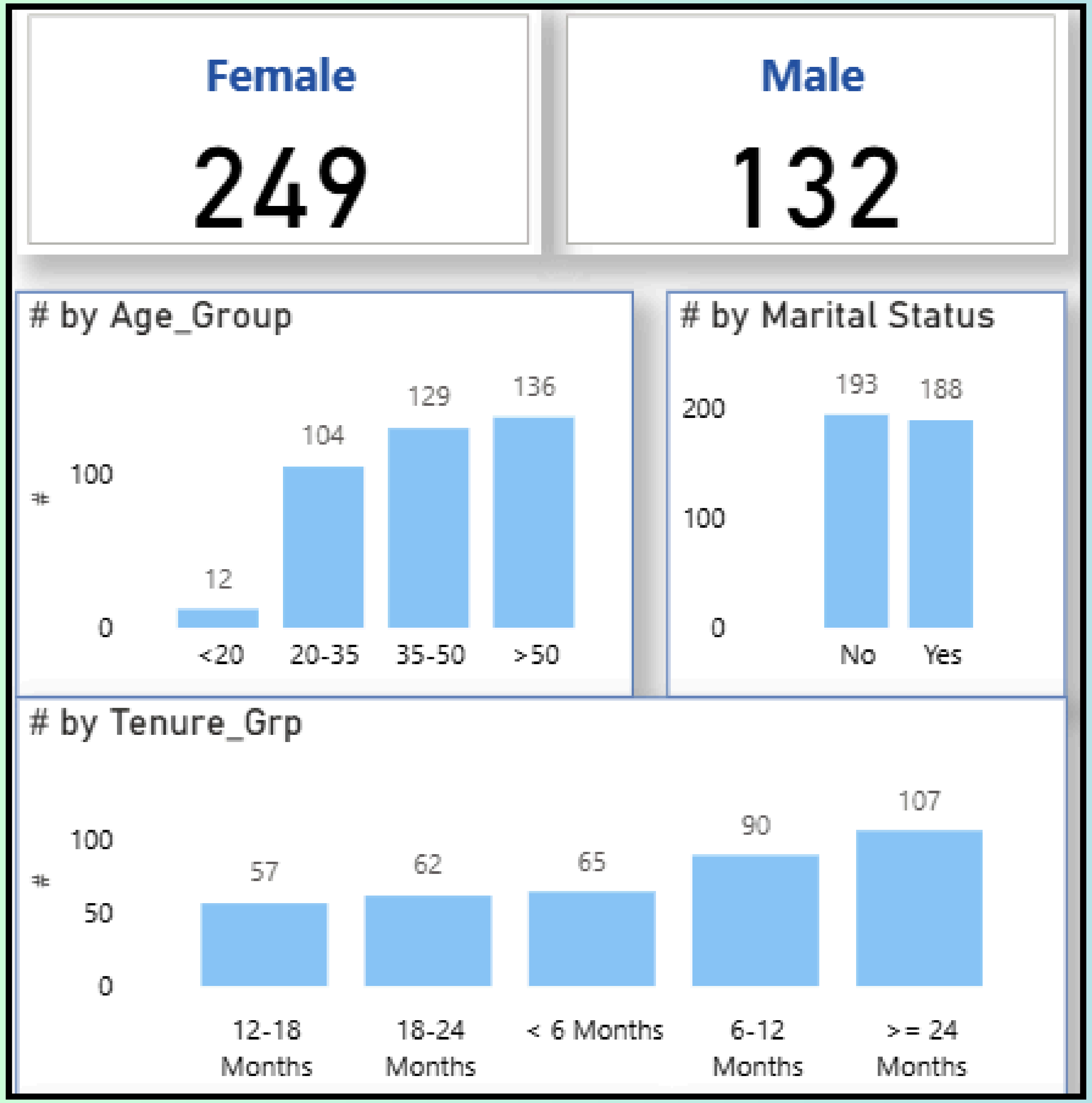
Churn Analysis - Prediction

Count of Predicted Churners : 381



Customer_ID	Sum of Monthly_Charge	Sum of Total_Revenue	Sum of Total_Refunds	Sum of Number_of_Referrals
79700-KER	107.95	456.10	0.00	5
84789-PUN	107.95	356.58	0.00	9
13123-BIH	100.20	253.62	0.00	13
23632-HAR	97.10	262.85	0.00	2
26902-TEL	95.85	98.55	0.00	7
13666-UTT	95.40	344.18	0.00	15
76419-KAR	94.90	282.11	0.00	14
97828-MAH	92.00	313.60	0.00	0
25517-RAJ	91.45	280.27	0.00	4
92494-MAH	91.15	272.06	0.00	11
34713-KER	90.75	229.09	0.00	1
83994-DEL	90.70	253.82	0.00	8
12056-WES	90.40	362.89	0.00	2
94070-JAM	90.10	114.43	0.00	8
19811-MAH	89.85	348.30	0.00	10
78220-TAM	89.85	319.75	0.00	14
96967-CHH	89.35	94.71	0.00	5
37131-MAH	89.25	105.89	0.00	6
78918-BIH	86.05	299.63	0.00	12
19998-AND	85.70	346.27	0.00	0
78253-GUJ	85.00	100.20	0.00	13
44700-DEL	84.60	91.94	0.00	10
66712-GUJ	84.30	276.21	0.00	6
73547-KAR	84.05	178.51	0.00	13
88111-MAH	83.40	126.46	0.00	15
21956-JHA	82.30	369.31	0.00	10
Total	**16,054.80**	**43,096.26**	**100.24**	**2773**

Prediction

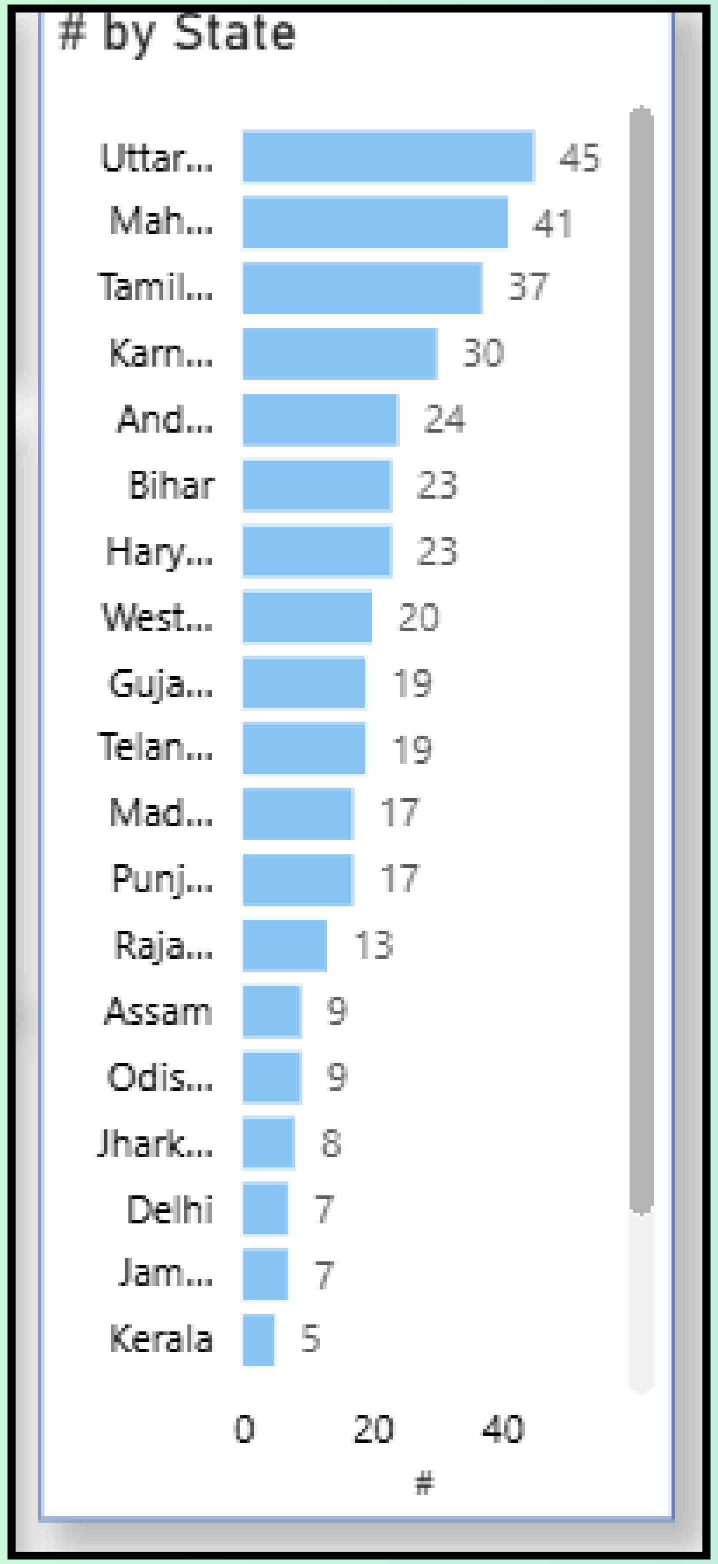


The dashboard predicts 381 churners, skewed toward females (249) over males (132).

Key segments

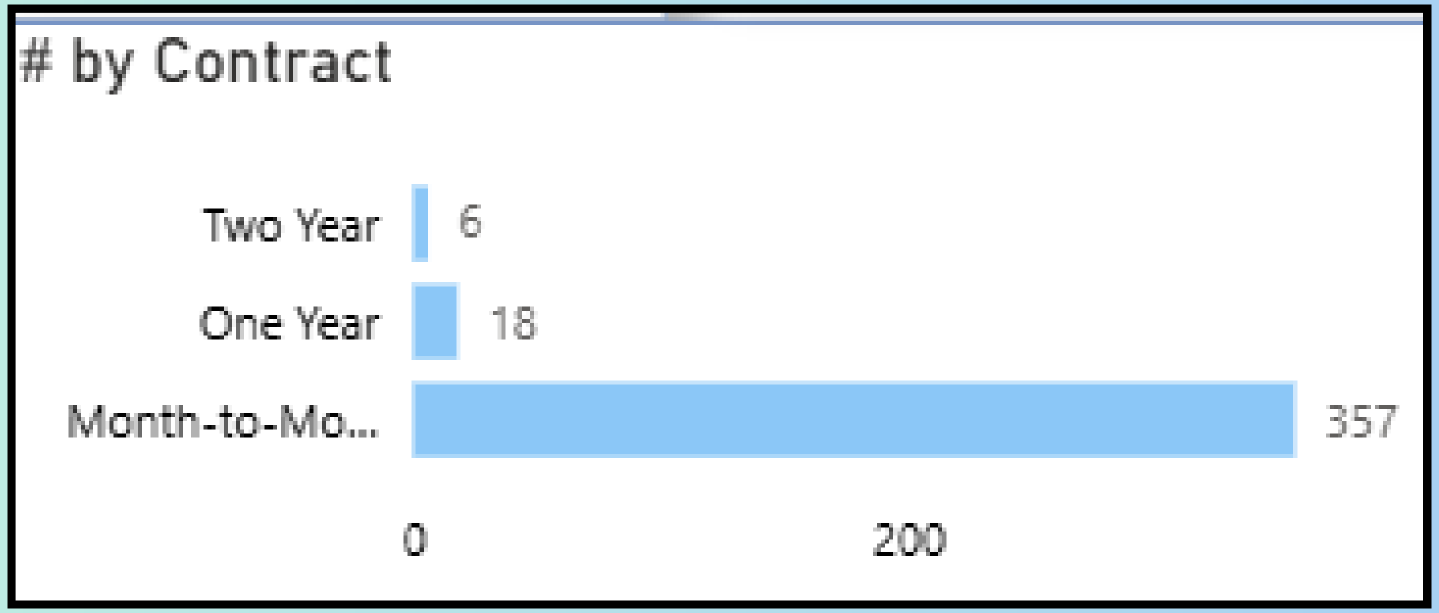
- ***Age:*** *Churners cluster in 35–50 (129) and >50 (136) groups, with very few under 20 (12).*
- ***Marital status:*** *Nearly even split between No (193) and Yes (188).*
- ***Tenure:*** *Higher counts at 6–12 months (90) and ≥24 months (107), with similar levels across 12–18 (57), 18–24 (62), and <6 months (65).*

Prediction



Geography and contracts

- Top states by churners include Uttar Pradesh (~45), Maharashtra (~41), and Tamil Nadu (~37), with others trailing.
- Contract type is dominated by Month-to-Month (357), with small counts for One Year (18) and Two Year (6).



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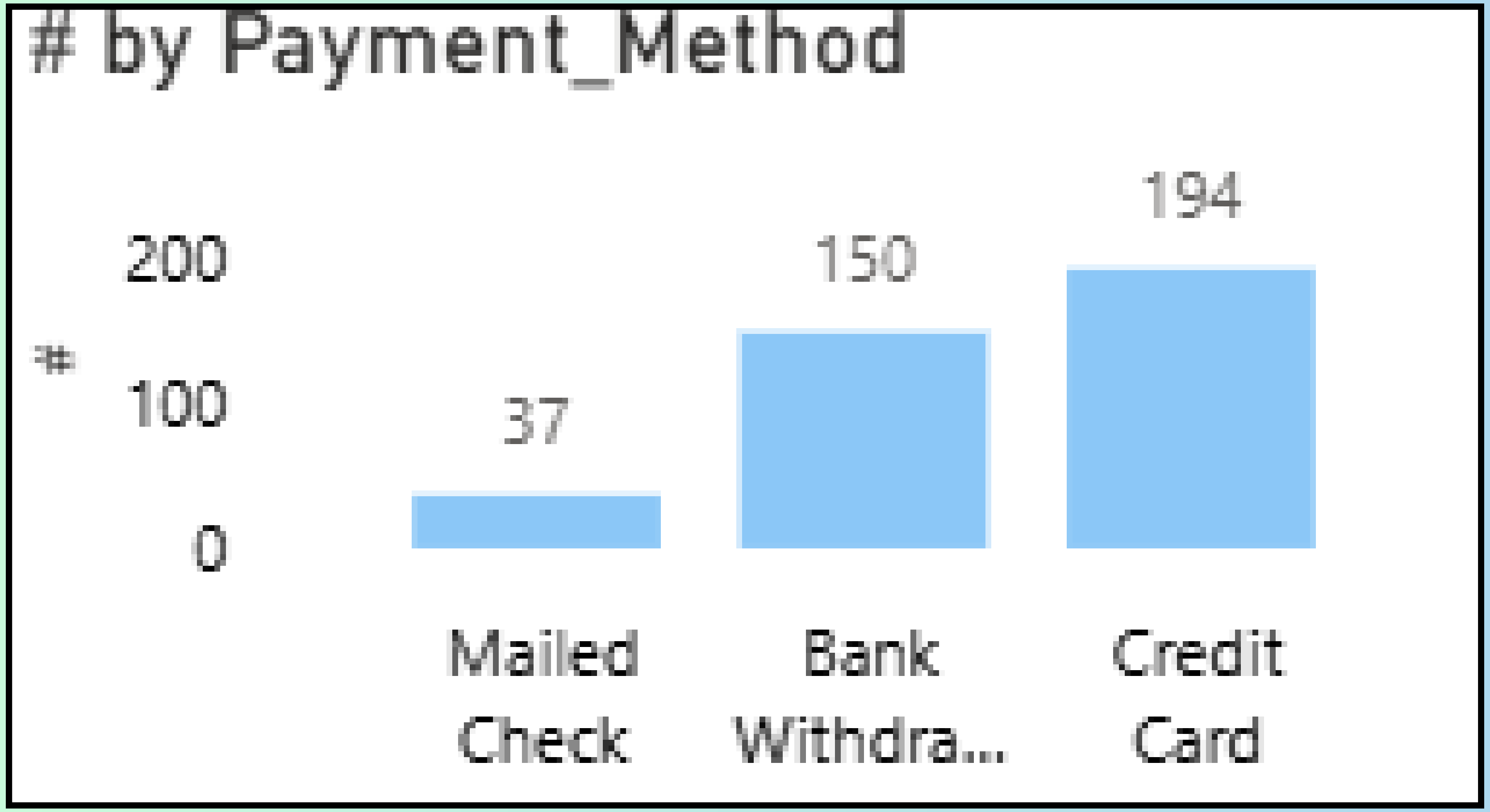

Payments and revenue

- ***Payment methods:*** Credit Card (194) is most common, followed by Bank Withdrawal (150) and Mailed Check (37).
- The listed churner table totals approximately ₹16,054.80 in monthly charges, ₹43,096.26 total revenue, 100.24 total refunds, and 2,773 referrals across the predicted churners shown.

Prediction Model

To predict customer churn, I applied Machine Learning models on the dataset.

- ***Key steps:***

Data Preparation

- Cleaned and preprocessed data (handled missing values, categorical encoding, scaling).
- Split dataset into training (70%) and testing (30%) sets.

Model Selection

- Tried different classification models such as:
- Logistic Regression – simple, interpretable baseline model.
- Random Forest – handled categorical features and gave good accuracy.

```
# Initialize the Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
# Train the model
rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

```
#Evaluate Model

# Make predictions

y_pred = rf_model.predict(X_test)
```

```
# Evaluate the model

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Confusion Matrix:
[[791  50]
 [125 236]]

Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.94      0.90       841
           1       0.83      0.65      0.73       361

    accuracy                           0.85      1202
   macro avg       0.84      0.80      0.81      1202
weighted avg       0.85      0.85      0.85      1202
```

Prediction Model

- ***Evaluation Metrics***
  - Measured performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC.
  - Example: Random Forest achieved ~82% accuracy with high recall for churn customers.
- ***Key Insights from Model***
  - Tenure, Contract Type, Payment Method, and State were the most important features for predicting churn.
  - Customers with month-to-month contracts, mailed checks, and short tenure are most likely to leave.
- ***Business Use***
  - The model helps identify at-risk customers early.
  - Companies can apply retention strategies (discounts, personalized offers, loyalty programs) for predicted churners.

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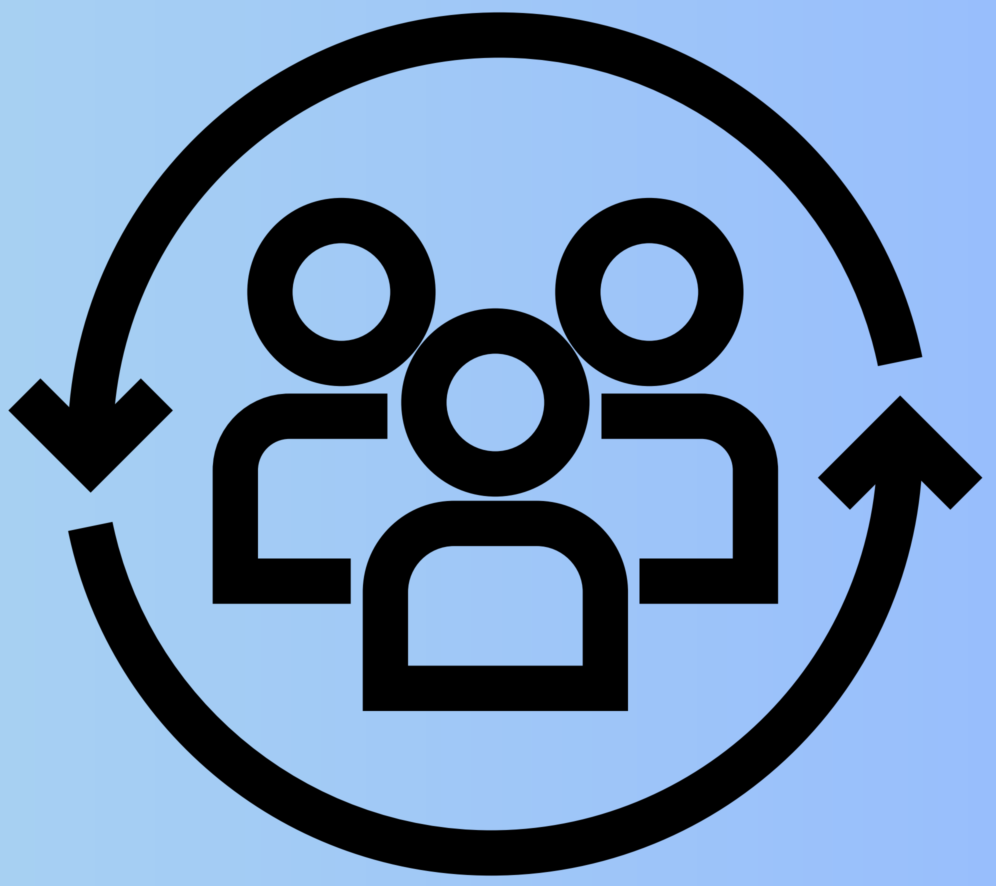
```

Conclusion

- Customer churn is significantly influenced by tenure, contract type, payment method, and geography.
- Short-term customers (month-to-month) and those using mailed checks show the highest churn rate.
- The predictive model (Random Forest) achieved good accuracy (~82%), effectively identifying customers at risk of leaving.
- Businesses can reduce churn by focusing on loyalty programs, flexible long-term contracts, and digital payment incentives.



Future Scope

1. **Model Enhancement**
- Use advanced models like Deep Learning (ANN, LSTM) for better accuracy.
- Apply ensemble methods to improve predictions further.
2. **Real-time Prediction**
- Integrate the churn model into CRM systems to predict churn in real-time.
3. **Personalized Retention Strategies**
- Build recommendation systems for targeted offers.
- Use customer segmentation to design different retention strategies.
4. **More Data Sources**
- Include social media, customer feedback, and transaction history to strengthen churn analysis.
5. **Cost-Benefit Analysis**
- Evaluate how much money is saved by retaining customers vs. acquiring new ones.

Thank you